PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Sex as a prognostic factor for mortality in critically ill adults with sepsis: a systematic review and meta-analys
AUTHORS	Antequera, Alba; Lopez-Alcalde, Jesus; Stallings, Elena; Muriel, Alfonso; Fernández Félix, Borja; del Campo, Rosa; Ponce Alonso, Manuel; Fidalgo, Pilar; Halperin, Ana Veronica; Madrid, Olaya; Álvarez, Noelia; Solà, Ivan; Gordo, Federico; Urrutia, Gerard; Zamora, Javier

VERSION 1 – REVIEW

REVIEWER	Wilcox, M. Elizabeth University of Toronto, Interdepartmental Division of Critical Care
REVIEW RETURNED	25-Feb-2021

GENERAL COMMENTS	Antequera and colleagues report their findings of a systematic review and meta-analysis to assess the role of sex as an independent prognostic factor in mortality after sepsis. They combined the results of 13 studies (n=80,520) and found no sex-based differences in all cause acute hospital or ICU mortality. The manuscript is very well written with clear study eligibility and inclusion criteria. It appropriately used a critical appraisal tool and conformed to PRISMA checklist. The authors clearly articulate their findings and limitations of their work including recognizing conceptual challenges existing in the literature between the terms sex and gender. Topic is highly relevant and timely. No major comments. Minor comments: 1. The use of “All cause hospital mortality (longest follow-up)” is confusing to the reader. Please revise or define (acute hospital mortality is a commonly used term – meaning in death while hospitalized is this what the authors mean?). 2. Would remove 7 day mortality when there is no reported data in any paper from Tables 2 and 8 as already busy. 3. Would consider removing Adrie et al paper from forest plot as no meta analysis made – please just report OR and CI in manuscript. Nice job!

REVIEWER	Scalea, TM University of Maryland Medical Center
REVIEW RETURNED	14-Mar-2021

GENERAL COMMENTS	The authors have conducted an exhaustive and detailed study analyzing the available literature as to whether sex is an independent prognostic factor for mortality in patients with sepsis admitted to the ICU. The strength of this manuscript is the huge numbers the authors analyzed. However, the quality of the data in each individual study is not clear. How stringent was the definition of sepsis? Were patients with sepsis, severe sepsis and/or septic shock all included and counted equally? Clearly, treatment protocols almost certainly differed between the institutions included in the studies the authors examined. There is no way for the authors to account for any potential confounding differences any of these may have had on their results. The authors cannot comment on cause of death. If all patients that died did so of sepsis, that is fine. However, I am certain that some of the deaths were from unrelated causes. The quality of the science here seems quite good. I believe the authors have done a very good job of study design and data analysis. . The authors found little evidence to conclude that sex is important in mortality from sepsis. I would ask the authors to address my concerns about additional limitations. After that is done, I believe this is appropriate to publish in BMJ Open

REVIEWER	Besen, Bruno Universidade de Sao Paulo Faculdade de Medicina, Medical Emergencies Discipline
REVIEW RETURNED	08-May-2021

GENERAL COMMENTS	Statistical Review: Overall issues: the authors designed and conducted a systematic review and meta-analysis about the prognostic impact of sex (female vs. male as reference) on mortality of ICU-admitted patients with sepsis. - A major issue is that the authors do not specify (in the methods session of the manuscript) what is their primary mortality outcome (hospital mortality or mortality at 28 days), although these were specified in the PROSPERO registration. Conditioning on specific time points is usually better to assess mortality, but pooling the longest follow-up mortality has been shown to be the best method in critically ill patient studies to reduce imprecision (because we do not always have the same outcome at the same time point to pool data), although there should be some differentiation between short term or medium/long-term outcomes. Ideally, the authors should consider all studies longest-follow-up to improve precision (10.1097/CCM.0000000000001631) and this should be their primary outcome. - Another issue is that the authors do not clearly specify the study design (It's a PICOTS, not only a PICO) they would search for. Statistical Issues:

	- The authors use the HKSJ method for random-effects meta-analysis, a valid and potentially better method than the D-L method. - The authors correctly don't present funnel plots, given the low number of studies for their primary outcome. - The authors present Tau2, I2 and prediction intervals to allow for heterogeneity evaluation, but only state that they evaluated heterogeneity with prediction intervals. Please revise. - The authors do not specify any methods to explore heterogeneity, except for subgroup analyses. - The presented figures can be more clinician-reader friendly. Presenting log-odds and standard errors in the forest plots is not the best practice since they are the same information as effect estimates (OR) and 95% CIs. The number of events and the total number of patients in each group should be presented, though. Minor issues: - Please briefly explain (and not only cite) what your refer to when stating "Phase 2" studies. I believe you refer to the PROGRESS framework.? - Please correct the figure legends throughout. For example: "Prospective nested case-control" and "Retrospective cohort" instead of lower case. - Please correct the figure X axis throughout to enhance interpretability. There is no good reason to go from 0.01 to 100 of odds ratio. It can be from 0.1 to 10 for example. Please, consider the data-ink ratio to make this decision. - Please check supplemental figure 2 prediction interval and either correct it or withdraw it (if it is correct) because as it is now it doesn't add information. Final Remarks: This seems to be a reasonably well conducted systematic review, but the meta-analysis step needs refinement in their presentation and as a reviewer I believe the choice (and presentation thereof) of the primary outcome was not the best and may net reconsideration. Thank you for the opportunity to review your manuscript.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1

Dr. M. Elizabeth Wilcox, University of Toronto

Comments to the Author:

Antequera and colleagues report their findings of a systematic review and meta-analysis to assess the role of sex as an independent prognostic factor in mortality after sepsis. They combined the results of 13 studies (n=80,520) and found no sex-based differences in all cause acute hospital or ICU mortality. The manuscript is very well written with clear study eligibility and inclusion criteria. It appropriately used a critical appraisal tool and conformed to PRISMA checklist. The authors clearly articulate their findings and limitations of their work including recognizing conceptual challenges existing in the literature between the terms sex and gender.

Topic is highly relevant and timely.

No major comments.

Minor comments:

1. The use of “All cause hospital mortality (longest follow-up)” is confusing to the reader. Please revise or define (acute hospital mortality is a commonly used term – meaning in death while hospitalized is this what the authors mean?).

Thank you, Dr Elizabeth Wilcox, for reading our piece of work critically. We appreciate your suggestions to enhance clarity of our manuscript.

We have modified the wording of primary outcomes for clarity. Now “All-cause hospital mortality” and “28-day all-cause mortality” are used, instead of “All-cause mortality (the longest follow-up provided by study authors)” and “28-day all-cause hospital mortality”, respectively. We have included Table 1 that displays the research question according to the PICOTS system for the sake of clarity.

Table 1. PICOTS system

Population	**Index prognostic factor**	**Comparator**	**Outcome(s)**	**Timing**	**Setting**
Adults with sepsis	Sex	Non-applicable to this review[1]	*Primary outcomes*		Intensive care units

	All-cause hospital mortality	The longest follow-up provided by the study authors (until death of hospital discharge)
	28-day all-cause mortality	28 days from sepsis diagnosis
	Secondary outcomes:	
	7-day all-cause hospital mortality	7 days from sepsis diagnosis
	1-year all-cause mortality	1 year from sepsis diagnosis
	All-cause ICU mortality	The longest follow-up provided by the study authors (until death of ICU discharge)

[1] Core set of adjustment factors**:** age, severity score [Sequential Organ Failure Assessment score (SOFA), Simplified Acute Physiology Score II (SAPS II) or Acute Physiologic Assessment and Chronic Health Evaluation II (APACHE II)], comorbidities (immunosuppression, pulmonary diseases, cancer, liver diseases, or alcohol dependence), non-urinary source of infection, and inappropriate or late antibiotic coverage.

We have noted this modification in the appendix (Supplemental Table 1) to ensure transparency regarding the protocol.

Extracted from Supplemental Table 1. Differences between the protocol and the review

Modified element	**Explanation**
Wording primary outcomes	We modify the wording for primary outcomes for clarity purposes, following the suggestion of peer reviewers. “All-cause hospital mortality” and “28-day all-cause mortality”, instead of All-cause mortality (the longest follow-up provided by study authors)” and 28-day all-cause hospital mortality”, respectively.

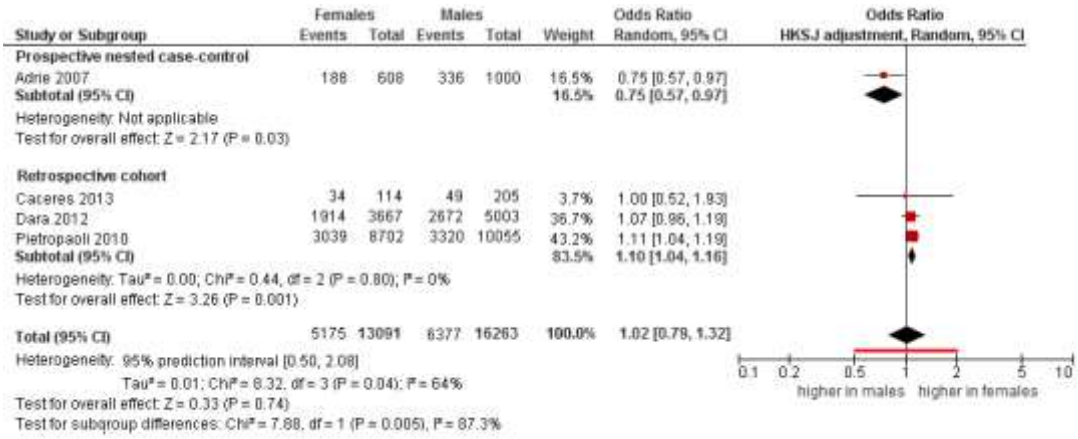
2. Would remove 7 day mortality when there is no reported data in any paper from Tables 2 and 8 as already busy.

Thank you for this suggestion, for clarity purposes, we have removed the column corresponding to 7-day mortality from Table 2 and Supplemental Table 8 as no study reported data.

3. Would consider removing Adrie et al paper from forest plot as no meta analysis made – please just report OR and CI in manuscript.

We understand how this might be misunderstood, but we think we think that it may be appropriate to keep Adrie et al in the forest plot of "All-cause hospital mortality" as its data were included in the meta-analyses. Adrie et al appears alone in the forest plot because it was the only study considered in that subgroup analysis (but, as stated above it was meta-analysed). Although we stated in the body of the manuscript that some subgroup analyses were not conducted because there were insufficient studies, we think the graphical visualisation helps identifying which subgroups were underrepresented (i.e., a single study prospective nested case-control, Adrie et al)

- Forest plot of unadjusted analyses for all-cause hospital mortality



Nice job!

Reviewer: 2

Dr. TM Scalea, University of Maryland Medical Center

Comments to the Author:

The authors have conducted an exhaustive and detailed study analyzing the available literature as to whether sex is an independent prognostic factor for mortality in patients with sepsis admitted to the ICU. The strength of this manuscript is the huge numbers the authors analyzed.

Thank you, Dr. TM Scalea, for assessing our manuscript and providing constructive feedback.

However, the quality of the data in each individual study is not clear. How stringent was the definition of sepsis? Were patients with sepsis, severe sepsis and/or septic shock all included and counted equally?

Thank you for raising the sepsis definition concerns. Despite the current international consensus definition for adult sepsis and septic shock (Sepsis-3) (2016), studies conducted before 2016 characterised their populations by prior definitions. We anticipated this potential heterogeneity, and the protocol outlined that we accepted the study authors' definition of sepsis and septic shock. We have now reported definitions provided by study authors in Supplemental Table 7 (sepsis-related terms and operational definitions), and mentioned this concern in the discussion:

"As well, clinical heterogeneity was substantial between the included studies, which differed regarding the sepsis definition used (i.e., diagnostic criteria, and sepsis and/or septic shock), illness severity measurements and score ratings, comorbidity burden, as well as in clinical practice (i.e., treatment protocols)".

Supplemental Table 7. Sepsis definition provided by the study authors

Study	Sepsis-related term for defining health condition	Operational definition
Adrie 2007*	Sepsis severe	Severe sepsis was defined as infection with two or more criteria for systemic inflammatory response syndrome and at least one criterion for organ dysfunction
Caceres 2013	Severe infection, hospital-acquired pneumonia	Severe infection was defined as hospital-acquired pneumonia, including ventilator-associated pneumonia and health-care associated pneumonia
Dara 2012	Sepsis shock	Non-provided
Luethi 2010	Septic shock	Septic shock was defined as two or more criteria for systemic inflammatory response syndrome and refractory hypotension (systolic blood pressure of b90 mmHg or a mean arterial pressure of 65 mmHg after an intravenous fluid challenge), or hyperlactatemia (blood lactate level of ≥4.0 mmol/L), or both.
Madsen 2014	Severe sepsis and septic shock	Severe sepsis or septic shock as defined by Surviving Sepsis Campaign.
Mahmood 2012	Sepsis	Non-provided
Nachtigall 2011	Sepsis	Sepsis, severe sepsis, and septic shock was defined according to the national and international sepsis guidelines, requering two or more criteria for systemic inflammatory response syndrome associated with an infection
Pietropaoli 2010	Severe sepsis and septic shock	Severe sepsis was defined as development of at least one severe acute organ dysfunction within 3 days of a presumed infection.
Sakr 2013	Severe sepsis	Sepsis syndromes were diagnosed according to the criteria proposed by the American College of Chest Physicians/ Society of Critical Care Medicine Consensus Conference [Severe sepsis: sepsis associated with organ dysfunction, hypoperfusion, or hypotension]
Samuelsson 2015	Sepsis	Non-provided
Sunden-Cullberg 2020	Severe sepsis and septic shock	Severe sepsis and septic shock were diagnosed using a modified version of the 1992 sepsis definition, in practice accepting a diagnosis of severe sepsis on the basis of infection plus organ dysfunction
van Vught 2017†	Sepsis	Sepsis was defined as an infection diagnosed with a "probable" or "definite" likelihood, plus at least one additional variable as described in the 2001 International Sepsis Definitions. Shock was defined by the use of vasopressors.
Xu 2019‡	Sepsis, severe sepsis and shock septic	Non-provided

Clearly, treatment protocols almost certainly differed between the institutions included in the studies the authors examined. There is no way for the authors to account for any potential confounding differences any of these may have had on their results.

Regarding the variability across treatment protocols, we prespecified a core set of covariates that included inappropriate or late antibiotic coverage. However, treatment variability across protocols of included studies was likely present but we were not able to describe in detail because of the poor reporting. We have emphasised this limitation in the discussion section:

"As well, clinical heterogeneity was substantial between the included studies, which differed regarding sepsis definition used (i.e., diagnostic criteria, and sepsis and/or septic shock), illness severity measurements and score ratings, comorbidity burden, as well as in clinical practice (i.e., treatment protocols)".

The authors cannot comment on cause of death. If all patients that died did so of sepsis, that is fine. However, I am certain that some of the deaths were from unrelated causes.

Concerning the causes of death, this information was unavailable. Following your suggestion, we have mentioned this issue as limitation:

"Another limitation is that we cannot provide information about the cause of death, particularly relevant for late mortality"

The quality of the science here seems quite good. I believe the authors have done a very good job of study design and data analysis.

We thank the reviewer for appreciating these issues.

The authors found little evidence to conclude that sex is important in mortality from sepsis. I would ask the authors to address my concerns about additional limitations. After that is done, I believe this is appropriate to publish in BMJ Open

Reviewer: 3

Dr. Bruno Besen, Universidade de Sao Paulo Faculdade de Medicina

Comments to the Author:

Statistical Review:

Overall issues: the authors designed and conducted a systematic review and meta-analysis about the prognostic impact of sex (female vs. male as reference) on mortality of ICU-admitted patients with sepsis.

Thank you Dr. Bruno Besen for reviewing our manuscript and your helpful comments.

- A major issue is that the authors do not specify (in the methods session of the manuscript) what is their primary mortality outcome (hospital mortality or mortality at 28 days), although these were specified in the PROSPERO registration.

We thank the reviewer for his comment. We agree in that the manuscript will benefit with a clearer statement in methods about our consideration of both outcomes (hospital mortality and mortality at 28 days) as co-primary outcomes [see section “Main outcome(s)] as it was pre-specified in PROSPERO and in the protocol published previously [see section “Main outcome(s), available at doi: 10.1136/bmjopen-2019-035927].

Eligibility criteria section now includes this reference and now reads:

“The co-primary outcomes were all-cause hospital mortality and 28-day all-cause mortality. Secondary outcomes were 7-day all-cause hospital mortality, 1-year all-cause mortality, and all-cause ICU mortality.”

Conditioning on specific time points is usually better to assess mortality, but pooling the longest follow-up mortality has been shown to be the best method in critically ill patient studies to reduce imprecision (because we do not always have the same outcome at the same time point to pool data), although there should be some differentiation between short term or medium/long-term outcomes. Ideally, the authors should consider all studies longest-follow-up to improve precision (10.1097/CCM.0000000000001631) and this should be their primary outcome.

Thank you for this interesting reference about outcomes in critically ill patients.

Regarding your concern about pooling the longest follow-up hospital mortality making a difference between short and medium/long follow-up, although four studies were meta-analysed for the outcome all-cause hospital mortality, only two of them reported the length of stay (Adrie 2007, Caceres 2013). Thus, we cannot perform additional analyses to explore potential differences, but we have mentioned this issue in the discussion, as follows:

"Albeit we meta-analysed studies providing all-cause hospital mortality to improve precision, additional analyses to explore potential differences between short and medium/long-term outcomes could not be performed because only two out of four included studies reporting the length of stay (43, 44)."

- Another issue is that the authors do not clearly specify the study design (It's a PICOTS, not only a PICO) they would search for.

Thank you for pointing out this issue. We have included Table 1 that displays the research question according to the PICOTS system.

Table 1. PICOTS system

Population	**Index prognostic factor**	**Comparator**	**Outcome(s)**	**Timing**	**Setting**
Adults with sepsis	Sex	Non-applicable to this review[1]	*Primary outcomes*		Intensive care units
			All-cause hospital mortality	The longest follow-up provided by the study authors (until death of hospital discharge)	
			28-day all-cause mortality	28 days from sepsis diagnosis	
			Secondary outcomes:		
			7-day all-cause hospital mortality	7 days from sepsis diagnosis	
			1-year all-cause mortality	1 year from sepsis diagnosis	

	All-cause ICU mortality	The longest follow-up provided by the study authors (until death of ICU discharge)	

[1] Core set of adjustment factors**:** age, severity score [Sequential Organ Failure Assessment score (SOFA), Simplified Acute Physiology Score II (SAPS II) or Acute Physiologic Assessment and Chronic Health Evaluation II (APACHE II)], comorbidities (immunosuppression, pulmonary diseases, cancer, liver diseases, or alcohol dependence), non-urinary source of infection, and inappropriate or late antibiotic coverage.

Statistical Issues:

- The authors use the HKSJ method for random-effects meta-analysis, a valid and potentially better method than the D-L method.

We thank the reviewer for appreciating this fact.

- The authors correctly don't present funnel plots, given the low number of studies for their primary outcome.

We thank the reviewer for appreciating this fact.

- The authors present Tau2, I2 and prediction intervals to allow for heterogeneity evaluation, but only state that they evaluated heterogeneity with prediction intervals. Please revise.

Thank you for point this. We have included these details in methods section.

"We also calculated I-squared and Tau-squared statistics to provide further quantifications of statistical heterogeneity."

We measured statistical heterogeneity computing prediction intervals, referring to the text and Table 3 - Summary of Findings, and graphically in the forest plots. Because of the widespread use of other statistics, such as I^2 and Tau, these data are also available at text and forest plots.

- The authors do not specify any methods to explore heterogeneity, except for subgroup analyses.

Thank you for your comment. We explored pre-specified sources of heterogeneity via subgroup analyses. We have slightly modified the corresponding sentence in the methods section for clarity purposes:

"*We examined statistical heterogeneity computing prediction intervals when the random-effects meta-analysis contained at least three studies (30,32). We planned to explore possible methodological causes of heterogeneity performing subgroup analyses* [methods section]"

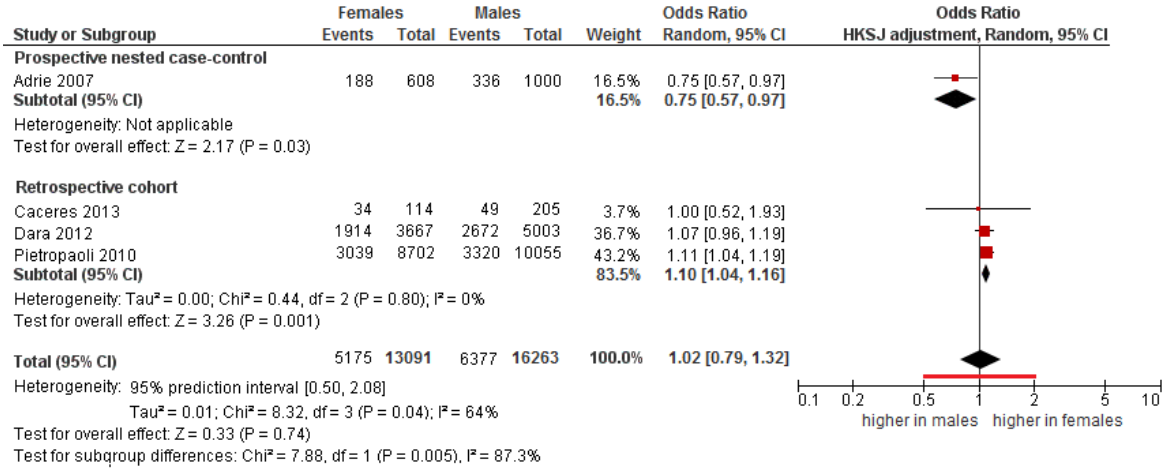
We have emphasised the presence of clinical heterogeneity in the discussion:

"As well, clinical heterogeneity was substantial between the included studies, which differed regarding sepsis definition used (i.e., diagnostic criteria, and sepsis and/or septic shock), illness severity measurements and score ratings, comorbidity burden, as well as in clinical practice (i.e., treatment protocols)".

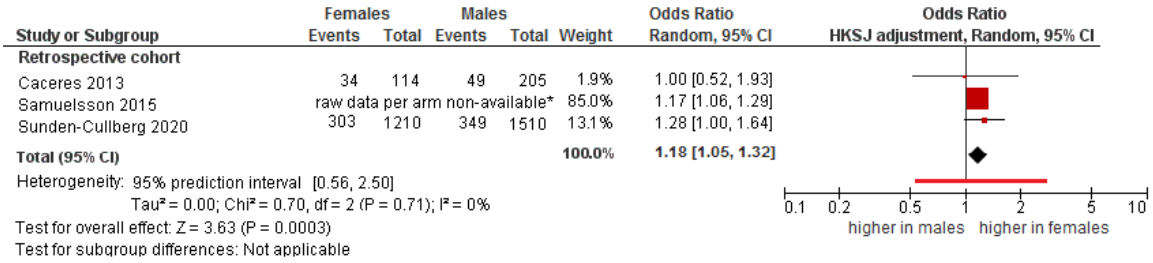
- The presented figures can be more clinician-reader friendly. Presenting log-odds and standard errors in the forest plots is not the best practice since they are the same information as effect estimates (OR) and 95% CIs. The number of events and the total number of patients in each group should be presented, though.

We depicted the results in the Summary of Findings that contains the absolute risk difference, which is considered the most effective manner of presenting results to clinicians (See doi: 10.1136/eb-2013-101340). Forest plots for adjusted estimates providing also the raw proportion of participants with the event per group may be misunderstood by the readers, as this review did not include RCTs for which the raw proportions are informative. Firstly, the proportions do not correspond with the adjusted estimate. Secondly, some studies provided adjusted estimates but no crude data (unadjusted, frequencies or proportions). However, we acknowledge that some readers may be interested in the raw data. Following your suggestion, we have adapted the forest plots for adjusted analyses and detailed the disaggregated raw data when available.

A. All-cause hospital mortality



B. 28-day all-cause mortality



* only provided the adjusted estimate

Minor issues:

- Please briefly explain (and not only cite) what your refer to when stating "Phase 2" studies. I believe you refer to the PROGRESS framework.?

Thank you for your comment. We have provided further details in the introduction:

"We included studies (experimental or any observational design) that sought to confirm the independent prognostic effect of sex on mortality in critically ill adults with sepsis controlling for covariates (called phase 2-confirmatory studies, which means the objective statement outlined sex as a prognostic factor of interest and analyses adjusted for covariates) [Hayden 2008, available at https://doi.org/10.1016/j.jclinepi.2007.08.005]"

- Please correct the figure legends throughout. For example: "Prospective nested case-control" and "Retrospective cohort" instead of lower case.

Thank you. We have modified it.

- Please correct the figure X axis throughout to enhance interpretability. There is no good reason to go from 0.01 to 100 of odds ratio. It can be from 0.1 to 10 for example. Please, consider the data-ink ratio to make this decision.

Thank you. We have modified it.

- Please check supplemental figure 2 prediction interval and either correct it or withdraw it (if it is correct) because as it is now it doesn't add information.

We confirm this prediction interval is right, and following your suggestion, we have removed its graphical representation.

Final Remarks:

This seems to be a reasonably well conducted systematic review, but the meta-analysis step needs refinement in their presentation and as a reviewer I believe the choice (and presentation thereof) of the primary outcome was not the best and may net reconsideration.

We have clarified the primary outcomes and formulated the research question according to the PICOTS system. We have addressed concerns arising from the meta-analyses presentation following your suggestions.

VERSION 2 – REVIEW

REVIEWER	Scalea, TM University of Maryland Medical Center
REVIEW RETURNED	26-Jun-2021

GENERAL COMMENTS	The authors have addressed my concerns. I feel this is now ready for publication

REVIEWER	Besen, Bruno Universidade de Sao Paulo Faculdade de Medicina, Medical Emergencies Discipline
REVIEW RETURNED	19-Jun-2021

GENERAL COMMENTS	The authors have addressed all statistical issues properly and should be congratulated for their manuscript.